Chapter Two Theory

CHAPTER TWO

TIME VALUE OF MONEY

2.1 Concept of Time Value of Money (TVM)

The Time Value of Money (TVM) is a fundamental financial principle stating that a dollar today is worth more than a dollar in the future due to its earning potential. This principle is based on the idea that money can earn interest over time, making its present value higher than the same nominal amount received in the future.

Key Reasons for TVM

Opportunity Cost - Money today can be invested to earn returns.

Inflation - Future money may have less purchasing power.

Risk & Uncertainty - Future cash flows are uncertain.

Consumption Preferences - People prefer money now rather than later.

TVM Formula Components

Present Value (PV) - The value of future money today.

Future Value (FV) - The value of present money in the future.

Interest Rate (i) - The rate at which money grows over time.

Number of Periods (n) - The time duration for which the money grows.

2.2 Simple Interest And Compound Interest

Simple Interest (SI)

Simple Interest is calculated only on the original principal amount.

Formula:

$$SI = P \times r \times t$$

Where:

- SI = Simple Interest
- P = Principal (Initial Amount)
- r = Annual Interest Rate (decimal form)
- t = Time (in years)

2.2 Simple Interest And Compound Interest

Compound Interest (CI)

Compound Interest is calculated on both the principal and the accumulated interest from previous periods.

Formula:

$$A = P \times (1 + r/n)^{n \times t}$$

Where:

- A = Final Amount
- P = Principal
- r = Annual Interest Rate (decimal form)
- n = Number of times interest is compounded per year
- t = Time (in years)

2.3 Economic equivalence

Economic equivalence is a concept used in financial analysis and engineering economics to compare different cash flows, investments, or financial decisions that may have different amounts, timing, or interest rates but are considered equal in value based on a given discount rate or time frame.

Key Aspects of Economic Equivalence:

Time Value of Money (TVM) - Money has different values at different points in time due to interest rates, inflation, and opportunity costs.

Present Worth (PW) and Future Worth (FW) - Different cash flows can be converted to a common point in time (present or future) to determine equivalence.

Annual Worth (AW) and Uniform Series - Cash flows can be converted into an equivalent uniform series (annuity) over a time period.

Interest Rates and Discounting - The choice of interest rate affects the equivalence calculation.

2.4 Nominal Rate of Interest (NIR)

The Nominal Interest Rate is the stated annual interest rate before considering compounding. It does not reflect the actual interest earned or paid in a year when compounding happens multiple times.

Formula:

$$r_{\text{nominal}} = n \times r_{\text{periodic}}$$

Where:

- r_{nominal} = Nominal annual interest rate
- n = Number of compounding periods per year
- r_{periodic} = Interest rate per compounding period

2.5 Effective Rate of Interest (EIR)

The Effective Interest Rate (EIR) or Annual Equivalent Rate (AER) accounts for the effect of compounding and reflects the actual interest earned or paid over a year.

Formula:

$$r_{\text{effective}} = \left(1 + \frac{r_{\text{nominal}}}{n}\right)^n - 1$$

Where:

- $r_{\text{effective}}$ = Effective annual interest rate
- r_{nominal} = Nominal annual interest rate (decimal form)
- n = Number of compounding periods per year

2.6 Continuous Compounding

Continuous compounding is the process where interest is added to the principal an infinite number of times within a given time period. It is the theoretical maximum that interest can compound, leading to the highest possible future value for a given interest rate.

2.7 Types of Cash Flow

a. Single Cash Flow

A single cash flow refers to a one-time receipt or payment of money at a specific point in time. This is the simplest form of cash flow and is often used in financial calculations such as the time value of money (TVM).

b. Uneven Payment Series

An uneven payment series refers to a cash flow pattern where payments (inflows or outflows) occur at different amounts over time, rather than being equal (like annuities). This type of cash flow is common in business projects, investments, and loan repayments.

2.7 Types of Cash Flow

c. Equal Payment Series

An equal payment series refers to a series of cash flows that are identical in amount and occur at regular intervals over a period of time. This type of cash flow is commonly found in:

Loans (e.g., mortgages, car loans)

Annuities (e.g., retirement payments, insurance payouts)

Savings plans (e.g., fixed monthly deposits)

d. Linear Gradient Series

A **linear gradient series** is a cash flow pattern where payments increase or decrease by a fixed amount each period. It is commonly used in:

Salary growth projectionsLoan repayment structuresMaintenance cost estimatesInvestment contributions with increasing deposits

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2.7 Types of Cash Flow

e. Geometric Gradient Series

A geometric gradient series is a cash flow pattern where payments increase or decrease by a fixed percentage (g%) each period, rather than a fixed amount. This type of cash flow is commonly used in:

Salary increases with percentage-based raisesInvestment growth with reinvested returnsInflation-adjusted costsLoan payments that increase by a fixed percentage

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